

3.3.8 Data compression

Content	Additional information
Explain what data compression is. Understand why data may be compressed and that there are different ways to compress data.	Students should understand that it is common for data to be compressed and should be able to explain why it may be necessary or desirable to compress data.
Explain how data can be compressed using Huffman coding. Be able to interpret Huffman trees.	Students should be familiar with the process of using a tree to represent the Huffman code.

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Be able to calculate the number of bits required to store a piece of data compressed using Huffman coding. Be able to calculate the number of bits required to store a piece of uncompressed data in ASCII.	Students should be familiar with carrying out calculations to determine the number of bits saved by compressing a piece of data using Huffman coding.
Explain how data can be compressed using run length encoding (RLE).	Students should be familiar with the process of using frequency/data pairs to reduce the amount of data stored.
Represent data in RLE frequency/data pairs.	Students could be given a bitmap representation and they would be expected to show the frequency and value pairs for each row, eg 0000011100000011 would become 5 0 3 1 6 0 2 1.